

Runs of homozygosity on the X chromosome partially resolve the *type* of first-degree incest in a female offspring

Working methods-note draft — v0.1. Numbers are from the simulations described below; references marked “verify” need a final citation check.

Summary

Autosomal runs of homozygosity (ROH) detected on SNP microarrays reliably reveal that a child’s parents are first-degree relatives, but cannot say *which* first-degree union (father–daughter, mother–son, or brother–sister) produced the child: all three give an autosomal inbreeding coefficient $F \approx \frac{1}{4}$. Because of the X chromosome’s sex-specific transmission, the X carries information the autosomes do not. Using gene-dropping simulation on empirical sex-specific genetic maps, we show that in a **female** offspring the expected X autozygosity is $\frac{1}{2}$ **for father–daughter and mother–son but only $\frac{1}{4}$ for brother–sister**, so the X-to-autosome homozygosity ratio cleanly separates sib–sib (≈ 1) from parent–child (≈ 2). Father–daughter versus mother–son, however, share the same expected X burden and differ only in ROH segment architecture; the Bayes-optimal accuracy for that pair from a single genome is ~ 0.68 – 0.71 . Detection is robust: because first-degree ROH are chromosome-scale, the X signal survives sparse markers and genotyping error. We note that this is an underused signal — clinical laboratories frequently *exclude* the X from homozygosity calculations — and that suitable clinical data (SNP arrays on girls born of incest) already exist.

Background

- SNP/chromosomal microarray is first-tier testing for developmental delay, intellectual disability, autism, and congenital anomalies, and incidentally reveals parental relatedness through genome-wide ROH.
- Incidental detection of first-degree incest on these arrays is documented, alongside the attendant consent, reporting, and psychosocial issues (Grote 2012, 2013; Tarini 2013; Helm 2013; Bennett 2021; Wang 2014; Fan 2013; Schaaf 2011).
- Standard practice estimates the *degree* of relatedness from autosomal ROH burden ($\sim 25\% \rightarrow$ first-degree) and, per ACMG guidance, deliberately does **not** assign the specific parental relationship, since array data is not a paternity test and ROH size maps

imperfectly to relationship (Fan 2013).

- The X chromosome is transmitted asymmetrically: a father passes his single X intact to every daughter (no recombination); a mother passes a recombined X; males carry only a maternal X. This asymmetry makes X autozygosity depend on the sexes *in the consanguinity loop*, not just its depth.

Theory

The X inbreeding coefficient of a female equals the X-kinship coefficient between her parents (McPeck framework). Evaluating the three first-degree pedigrees:

Union (female offspring)	Autosomal F	Expected X F (F_X)
Father × daughter	$\frac{1}{4}$	$\frac{1}{2}$
Mother × son	$\frac{1}{4}$	$\frac{1}{2}$
Brother × sister	$\frac{1}{4}$	$\frac{1}{4}$

Consequences:

- The autosomes are blind to type (all $\frac{1}{4}$).
- The **F_X / F_auto ratio** is ≈ 2 for parent–child and ≈ 1 for sib–sib → separates sib–sib from the parent–child pair.
- Father–daughter and mother–son share $F_X \approx \frac{1}{2}$ and differ only in segment structure: father–daughter has one recombination-free X copy (the intact paternal X), so its autozygous tracts are bounded by a *single* maternal meiosis; mother–son’s two copies are separated by *two* meioses.

Methods

Gene-dropping simulation. Founder X (and autosomal) haplotypes were assigned unique labels and dropped through each pedigree with sex-specific recombination: female meioses used the empirical female map (Bhérier 2017 refined map, GRCh37; cross-checked against deCODE/Haldorsson 2019, GRCh38), males transmitted the X intact, and autosomes recombined in both sexes using sex-specific maps. Crossovers were drawn as a Poisson process in genetic distance (Haldane, no interference). Autozygosity was scored as the genome fraction where both copies carried the same founder label. PAR1/PAR2 were excluded (consistent with the maps). Female X genetic length ≈ 176 cM ($\lambda \approx 1.76$ crossovers per female meiosis); the empirical female–X map and the deCODE chrX map agreed on X

length to within ~ 0.4 cM.

Realistic genotype layer. SNP panels at array-like ($\sim 12k$) and WGS-like ($\sim 50k$) density across the X were simulated with a common-variant allele-frequency spectrum; founder haplotype alleles were dropped through the pedigree to produce offspring genotypes, with 0.5% genotyping error injected as false heterozygotes. ROH were called with a standard PLINK-style sliding-window algorithm (window 50 SNPs, ≤ 2 heterozygotes, ≥ 1.5 Mb, ≥ 50 SNPs).

Bayes ceiling (father–daughter vs mother–son). The number of IBD/non-IBD junctions on the X follows Poisson(λ) for father–daughter (one maternal meiosis) and Poisson(2λ) for mother–son (two female meioses). Optimal accuracy from the junction count is $\frac{1}{2} \cdot \sum_k \max[\text{Poisson}(k; \lambda), \text{Poisson}(k; 2\lambda)]$. A simulation classifier (gradient boosting, 5-fold CV) trained on full features (segment count, F_X , longest segment, length spread) gives the full-information ceiling.

Results

- **Means match theory** (validation): $F_X = 0.50 / 0.50 / 0.25$ for father–daughter / mother–son / sib–sib, and $F_{\text{auto}} \approx 0.25$ for all three (outbred control 0). (*Figure 2*)
- **Ratio separates sib–sib:** $F_X/F_{\text{auto}} \approx 2.0$ (father–daughter), 2.1 (mother–son), 1.0 (sib–sib). (*Figure 3, panel 1*)
- **Detection is robust:** detected F_X tracked true F_X to within ~ 0.001 at both densities despite genotyping error — first-degree ROH are chromosome-scale, so the limit is statistical, not technical. (*Figure 3, panel 2*)
- **Father–daughter vs mother–son is the wall:** equal F_X ; mean detected ROH segment counts ≈ 1.3 vs ≈ 2.1 . Bayes accuracy from junction count = **0.71**; full-information classifier = **0.68** (chance = 0.50); segment lengths add essentially nothing beyond the count. (*Figure 3, panel 3*)

Net: the X turns “which first-degree union?” into one easy classification (sib–sib vs parent–child) and one near-intractable one (father–daughter vs mother–son) from a single female genome.

Real-world anchor cases

Three published clinical cases anchor the simulation in girls actually born of incest:

- **Fan et al. 2013, Case 001** — a 6-year-old girl, ~ 598 Mb of homozygosity ($F \approx \frac{1}{4}$), referred for intellectual disability; parents a **brother–sister** pair; later given a tentative

diagnosis of α -ketoglutarate dehydrogenase deficiency uncovered through the homozygosity. A sib-sib girl — predicted lower X burden ($F_X \approx 1/4$); her X was not specifically flagged.

- **Fan et al. 2013, Case 004** — a 23-day-old female neonate with congenital anomalies, 345 Mb homozygosity ($F \approx 1/8$); parents an **uncle-niece** pair (second-degree).
- **Sund et al. 2013** — among 25 female patients in whom the X was evaluated, one **girl with a suspected first-degree relationship had an entirely homozygous X**, raising total homozygosity by ~4% and supporting the first-degree call. A fully homozygous X ($F_X \approx 1$) is the high tail that essentially only parent-child unions reach — the opposite end of the distribution from Fan's sib-sib girl.

These two girls bracket the model: a sib-sib girl with low X burden and a parent-child girl with a fully homozygous X. Notably, Sund et al. recommended that laboratories consider **excluding** the X from homozygosity calculations — i.e., the one chromosome carrying the type-discriminating signal is the one routinely discarded.

The gap

No published work appears to use X-ROH to discriminate the *type* of first-degree union. The X-kinship theory exists but has not been applied this way; clinical reports use autosomal ROH for the *degree* and (per ACMG) avoid assigning the relationship; forensic X-STR and machine-learning work treat the X as a complementary *detector* of incest and omit mother-son. The proposal here — read the X-ROH pattern in a female offspring to separate sib-sib from parent-child, and to bound how far father-daughter vs mother-son can ever be pushed — is, to our knowledge, unaddressed.

Limitations

- The SNP genotypes are simulated at realistic density rather than drawn from a specific reference panel (1000G VCFs were not accessible in the compute environment); the detection conclusions concern density and het-noise, which this reproduces.
- Crossovers are modeled without interference (Haldane); adding a chiasma-interference model (e.g., the map's native interference) would slightly tighten segment counts.
- PAR1/PAR2 are excluded.
- The accuracy bounds are for a **single** female genome. In casework, a genotyped parent or relative would likely break the father-daughter/mother-son degeneracy via the intact-paternal-X check — the obvious next experiment.

- Heightening the resolution from “first-degree” to a specific union raises the reporting and consent stakes; any application must engage the ELSI literature (Tarini 2013; Grote 2012/2013; Bennett 2021).

Conclusion

The X chromosome adds genuine, quantifiable information about the type of first-degree incest in a female offspring: it cleanly separates brother–sister from parent–child via a ~2-fold difference in X autozygosity that is robust to realistic genotyping, while father–daughter versus mother–son remains near a coin flip from a single genome (~0.68–0.71 ceiling). The signal is currently discarded in clinical practice, and the data to test it — SNP arrays on girls born of incest — already exist.

Figures

- **Figure 1.** Female recombination landscape of chrX (Bhérier refined map), with PAR1 excluded, the pericentromeric cold valley, and the distal rise toward PAR2. (*chrX_female_recomb.png*)
- **Figure 2.** Distribution of X autozygosity F_X by pedigree (means 0.50 / 0.50 / 0.25 / 0), and the father–daughter vs mother–son segment-count contrast. (*Fx_incest_sim.png*)
- **Figure 3.** Three-panel summary: (1) F_X/F_{auto} ratio splits sib–sib from parent–child; (2) detected vs true F_X (detection $\approx$ truth); (3) Poisson junction-count likelihoods and the father–daughter vs mother–son Bayes ceiling. (*incest_X_summary.png*)

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